

Model Analysis Report

Generated: March 11, 2026

Dataset Information

Number of timesteps: 136214

Spatial grid: 50 latitudes × 24 MLTs

Time range: 2019-11-01 00:00:00 to 2024-01-31 23:56:00

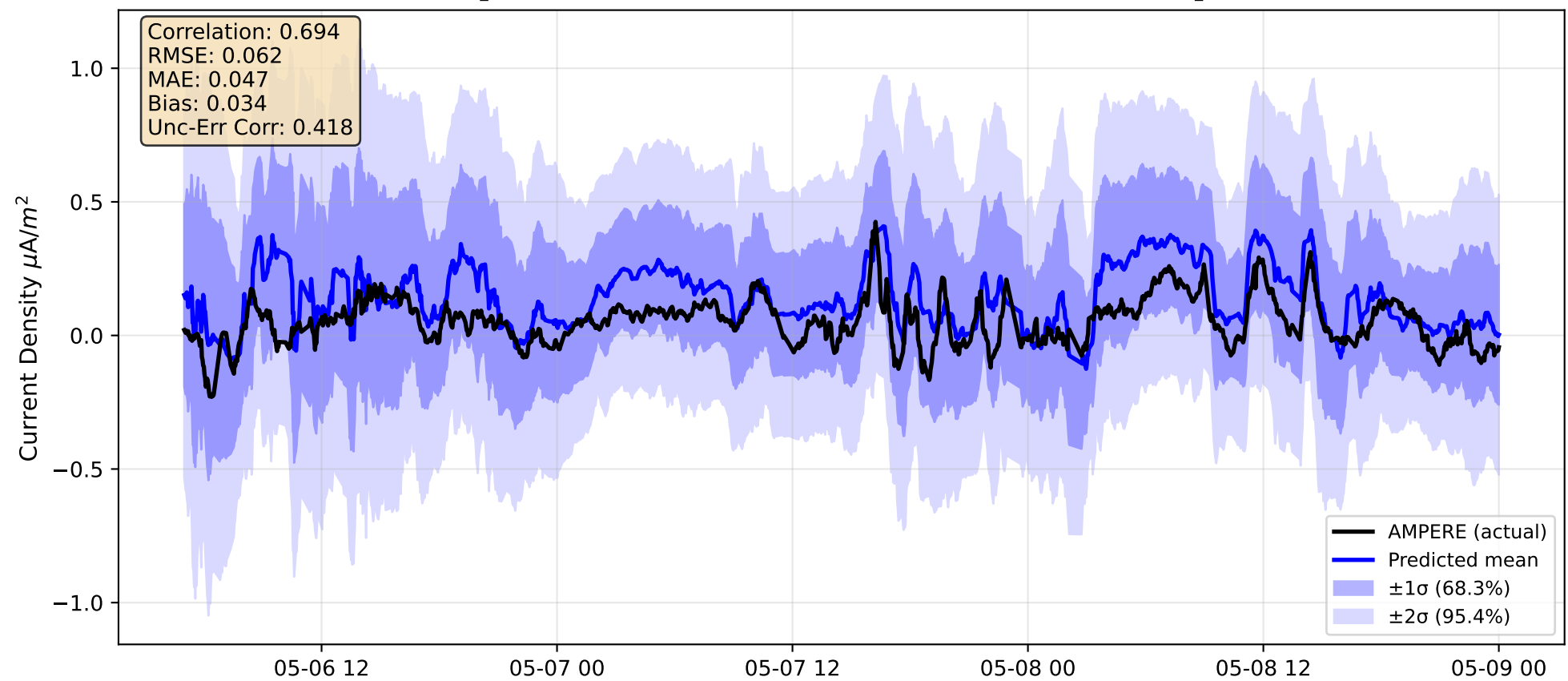
Uncertainty available: Yes

Analysis location: Lat 75-85°, MLT 15-21

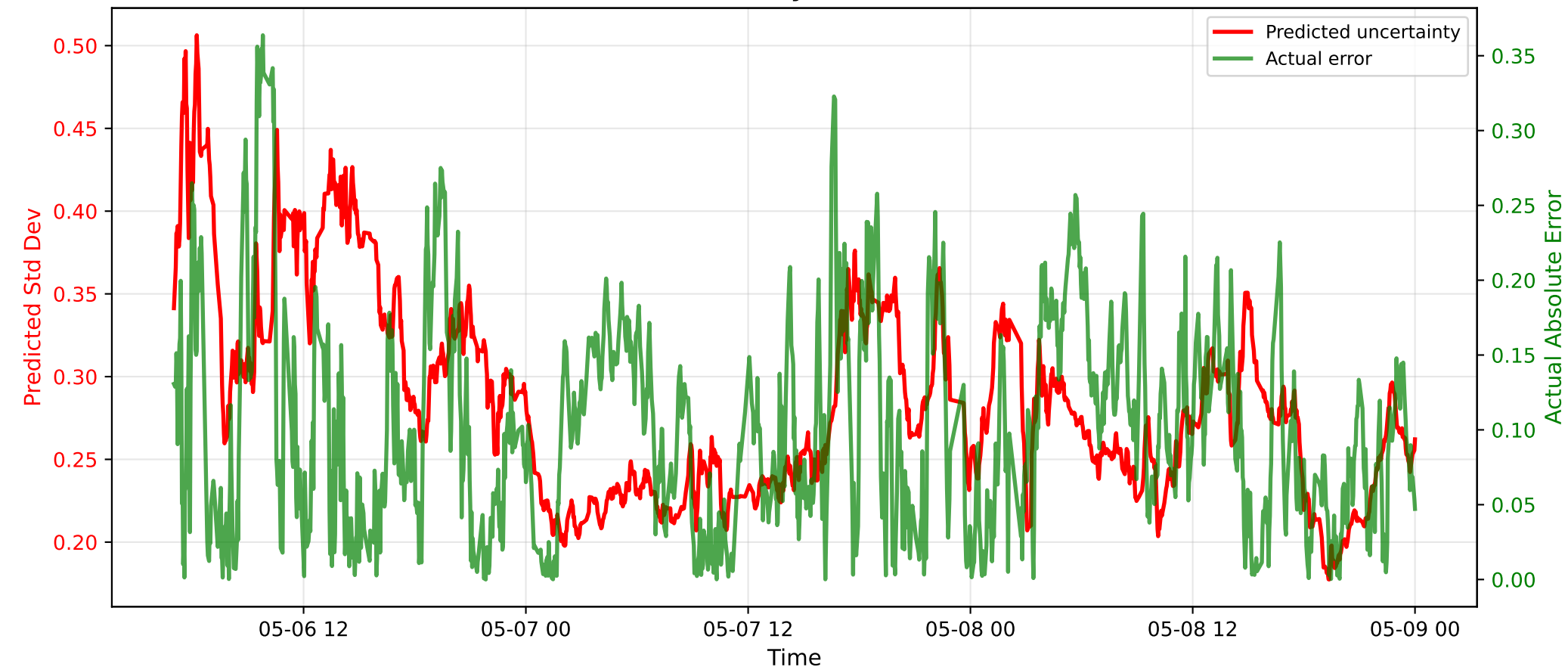
Plot time range: 2023-05-04 00:00:00 to 2023-05-09 00:00:00

*This report contains comprehensive validation metrics
for geomagnetic model predictions*

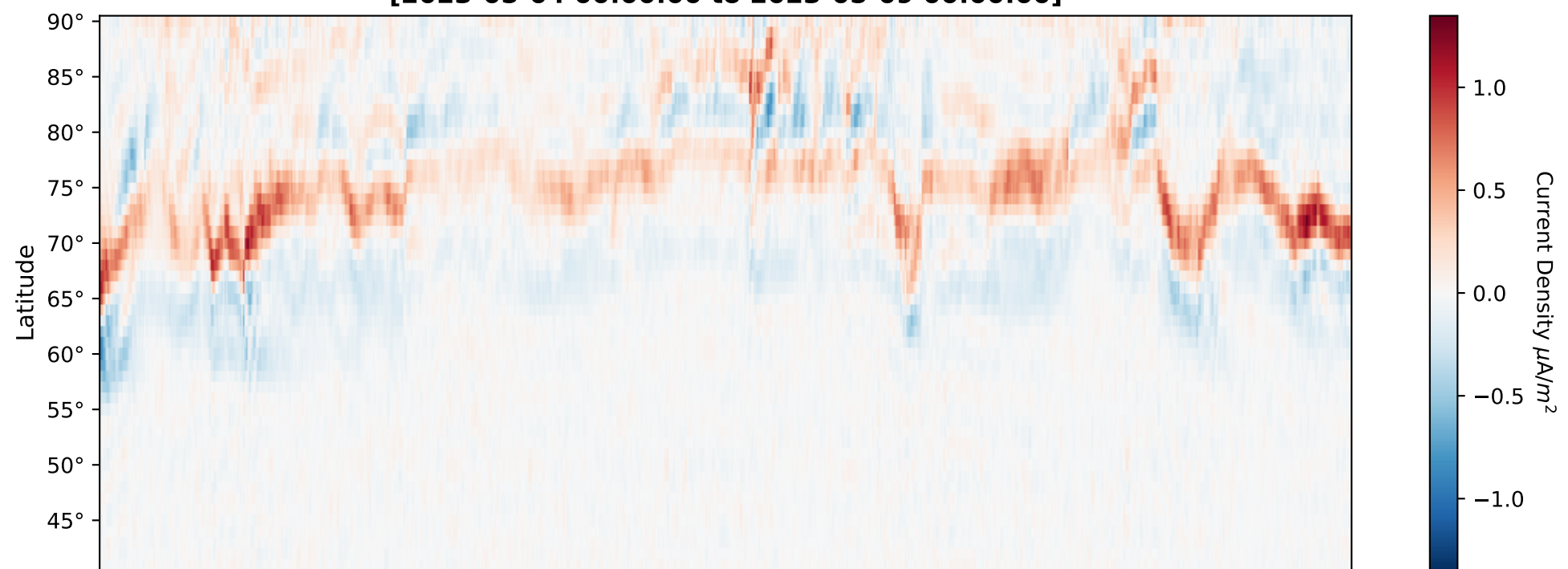
Time Series with Uncertainty (Lat 75-85°, MLT 15-21)
[2023-05-04 00:00:00 to 2023-05-09 00:00:00]



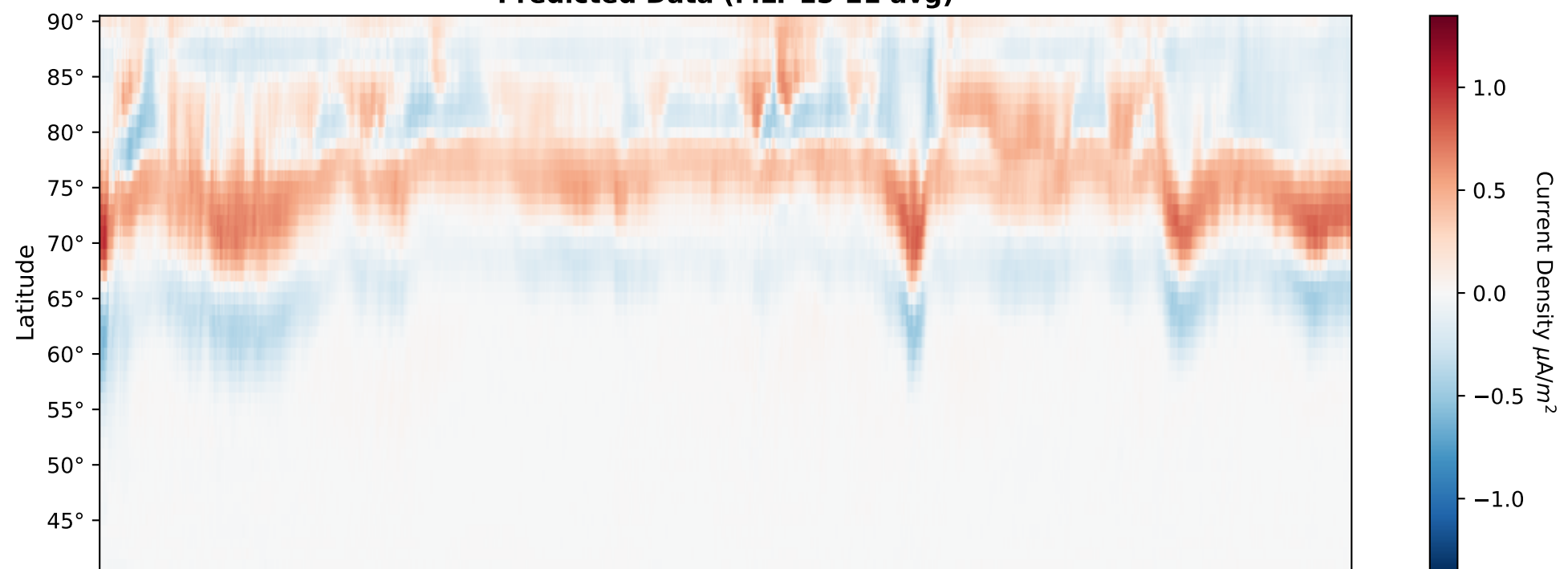
Uncertainty Evolution



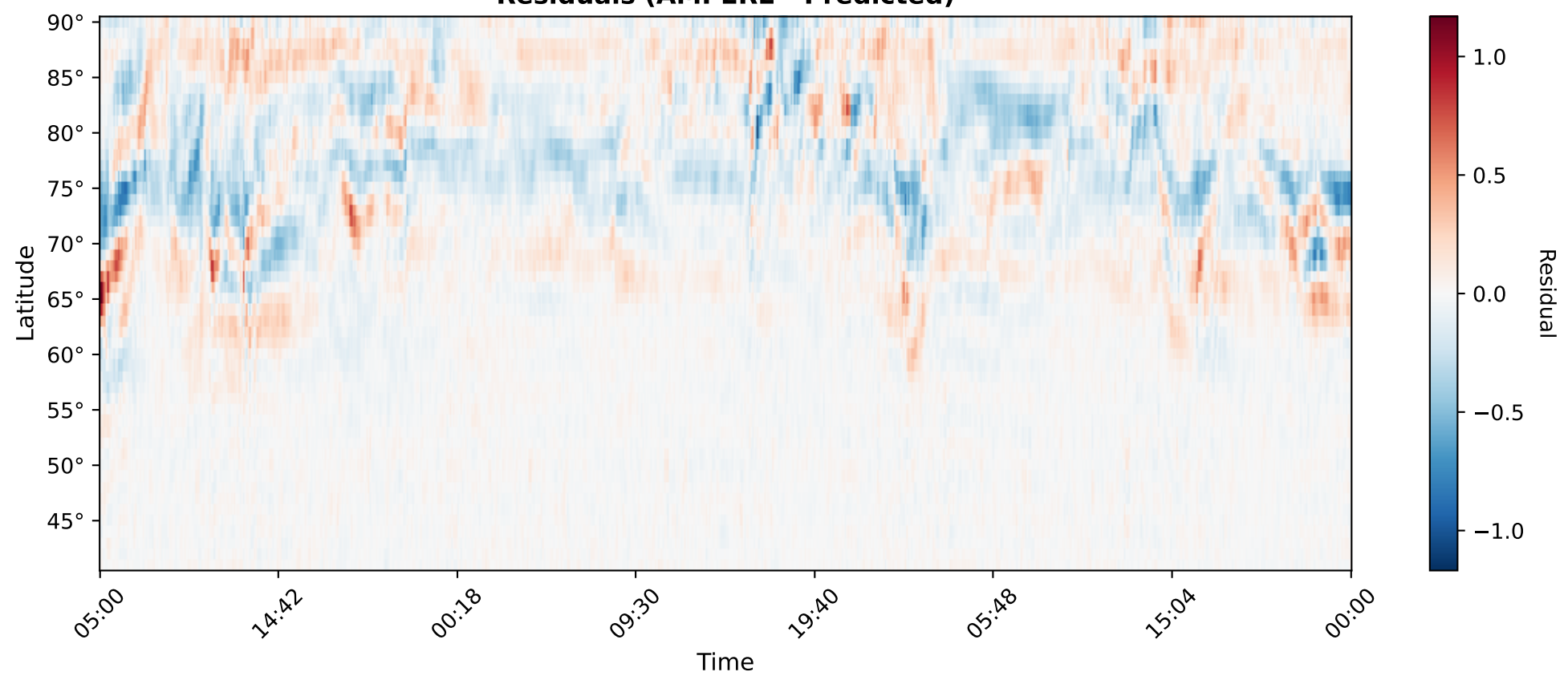
AMPERE Data (MLT 15-21 avg)
[2023-05-04 00:00:00 to 2023-05-09 00:00:00]



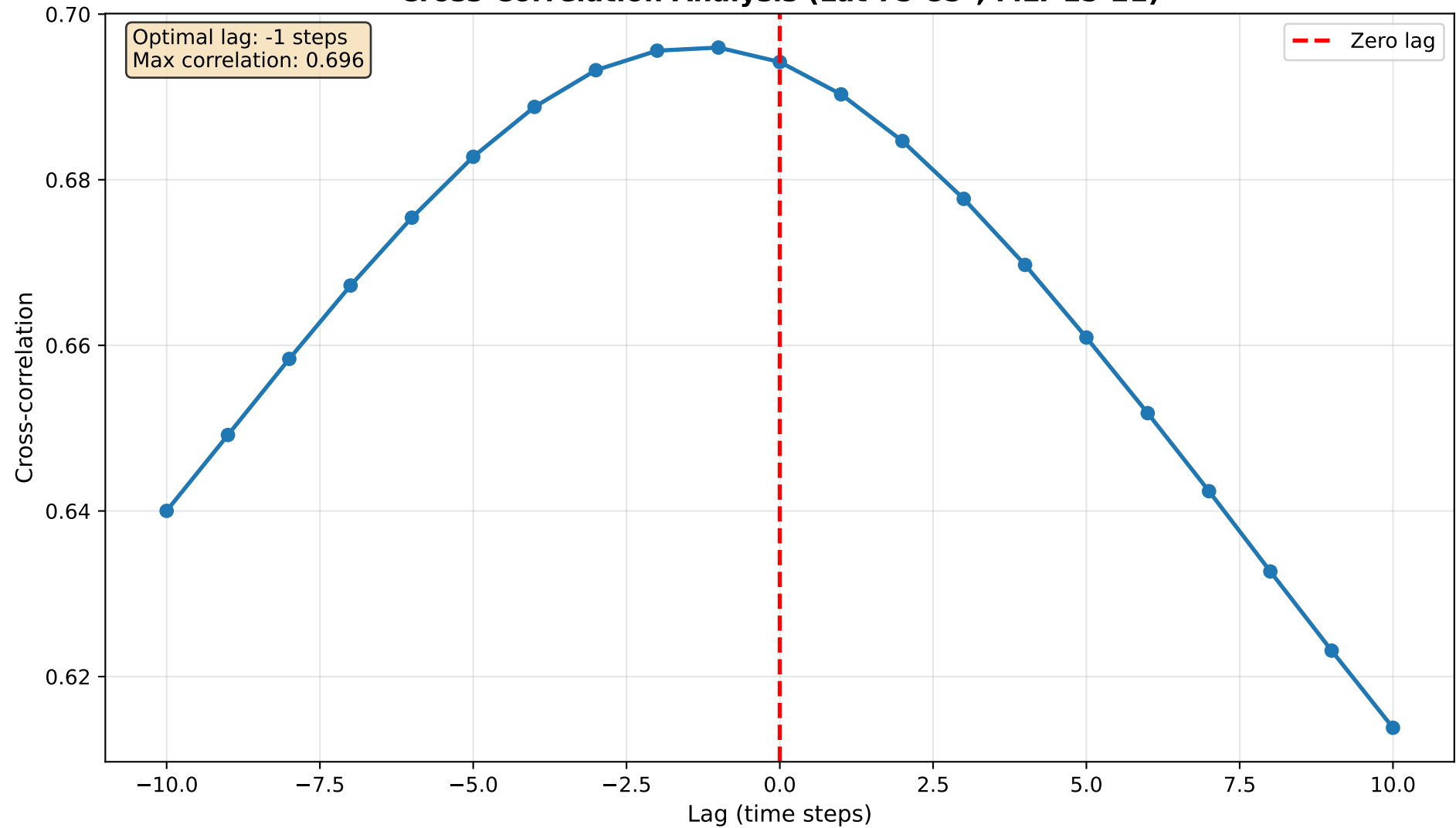
Predicted Data (MLT 15-21 avg)



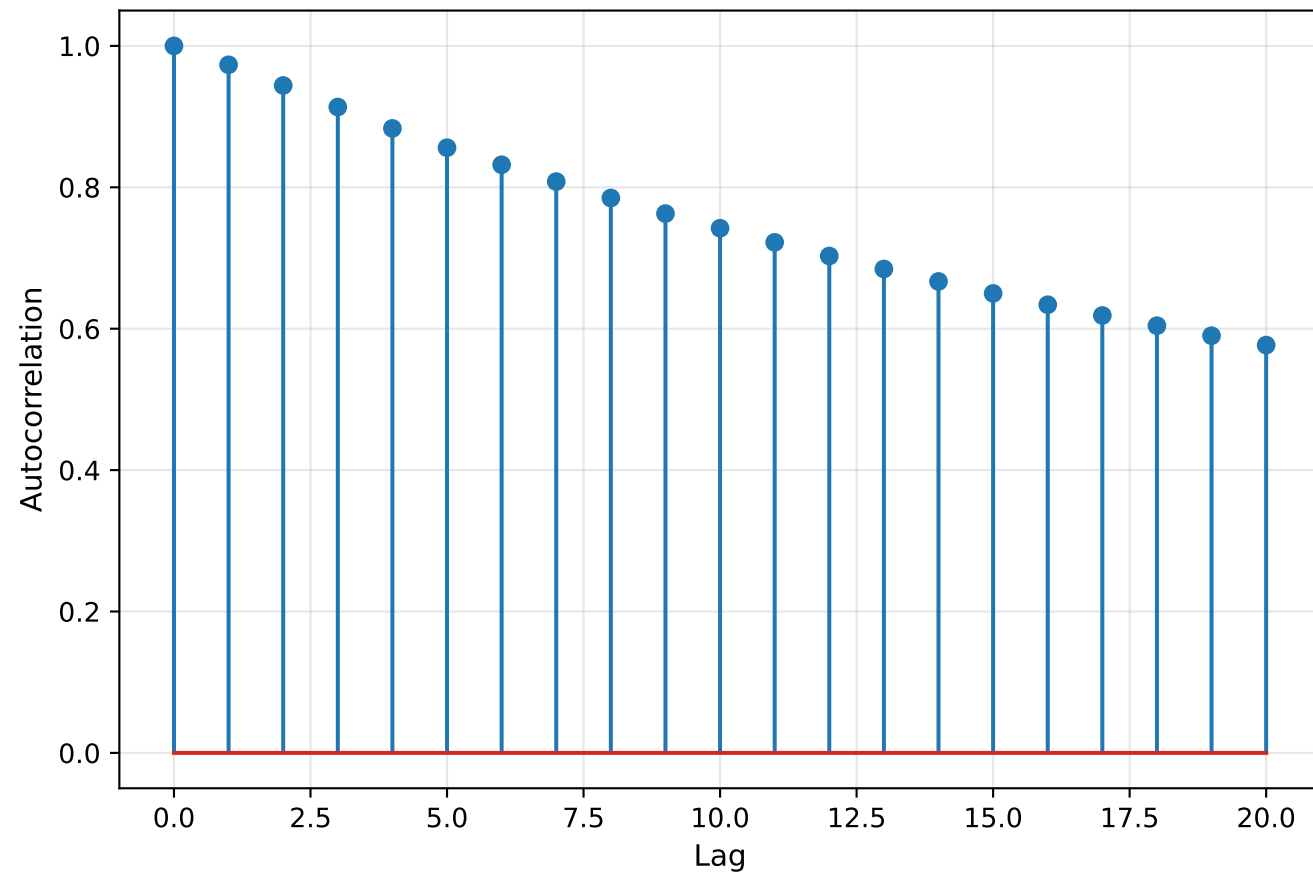
Residuals (AMPERE - Predicted)



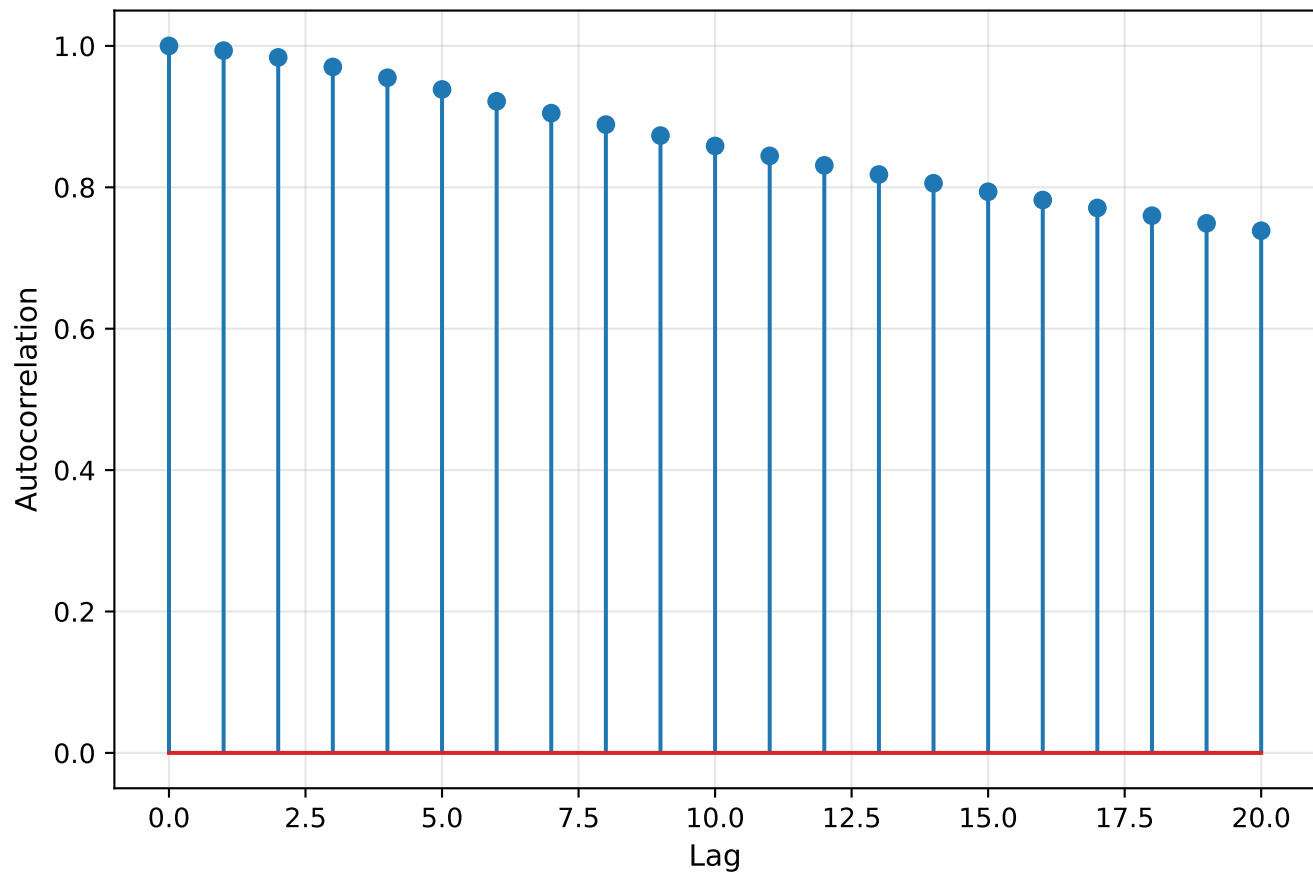
Cross-Correlation Analysis (Lat 75-85°, MLT 15-21)

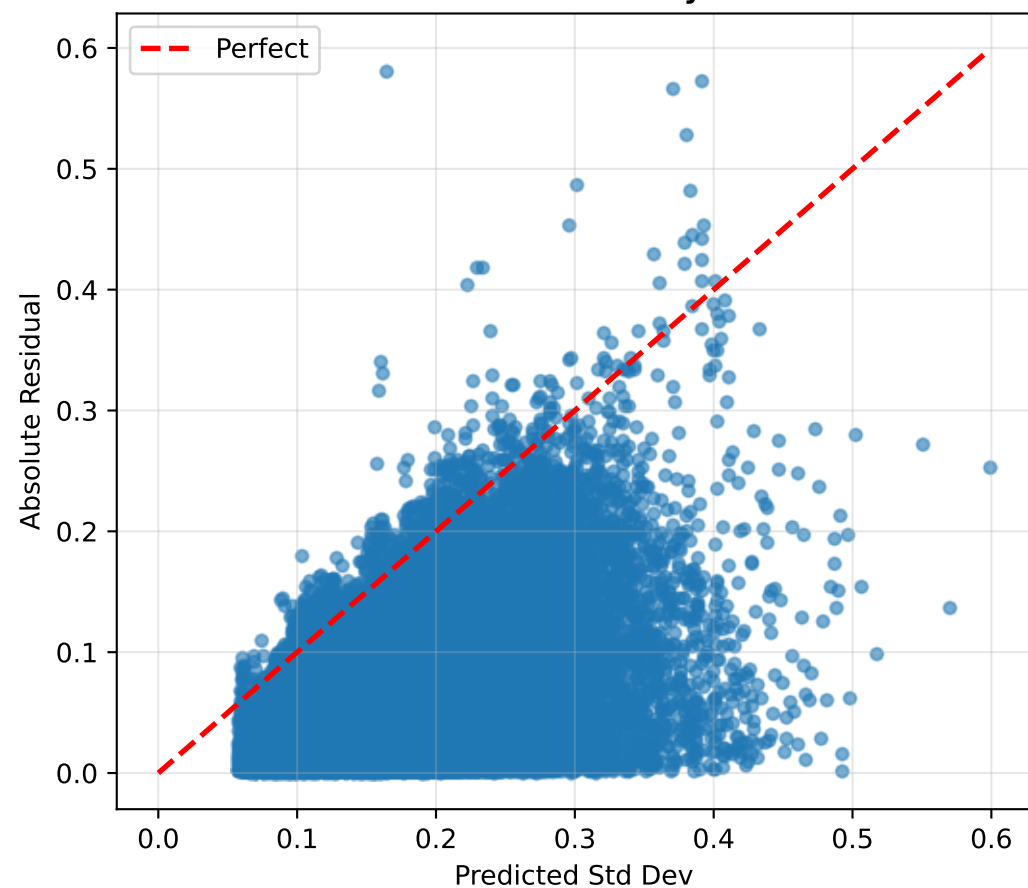
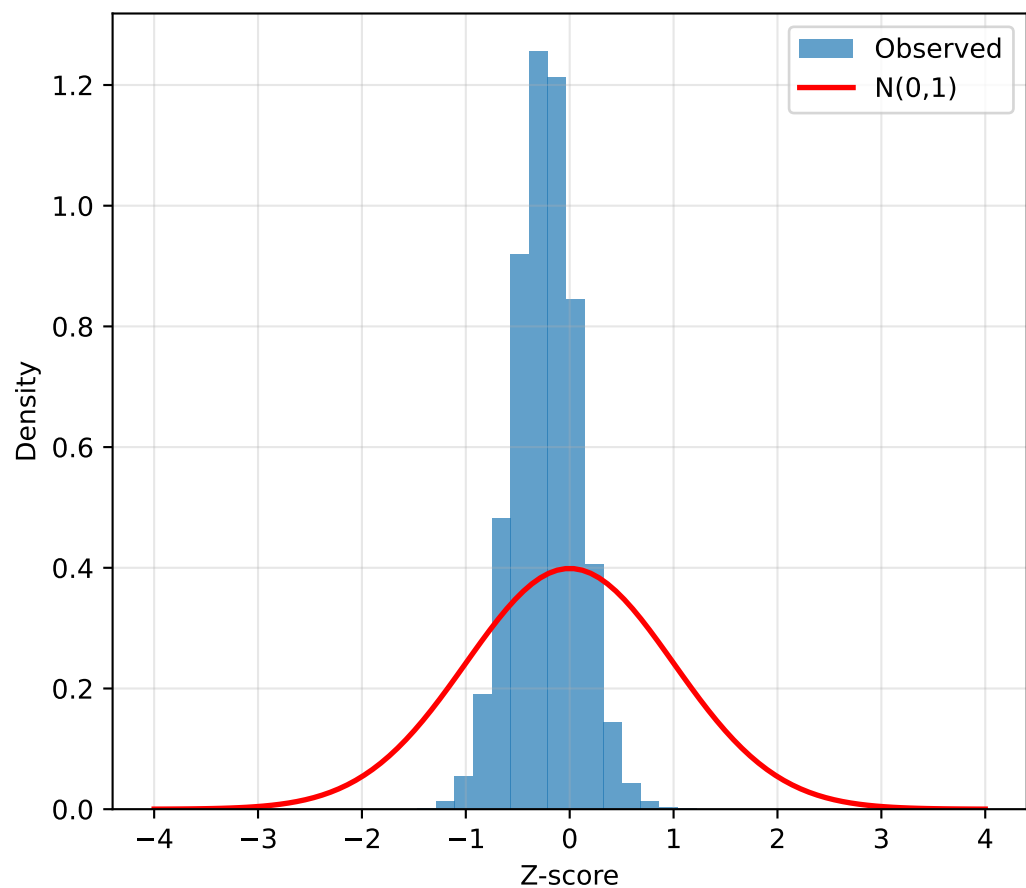
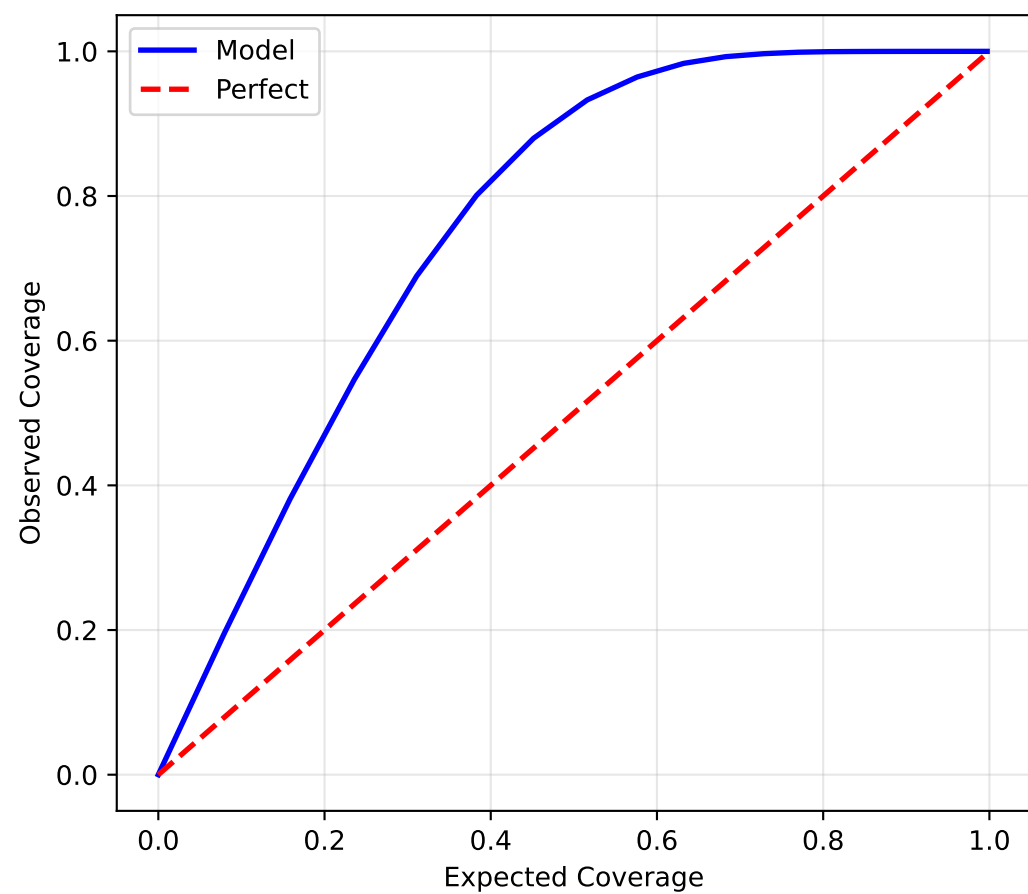
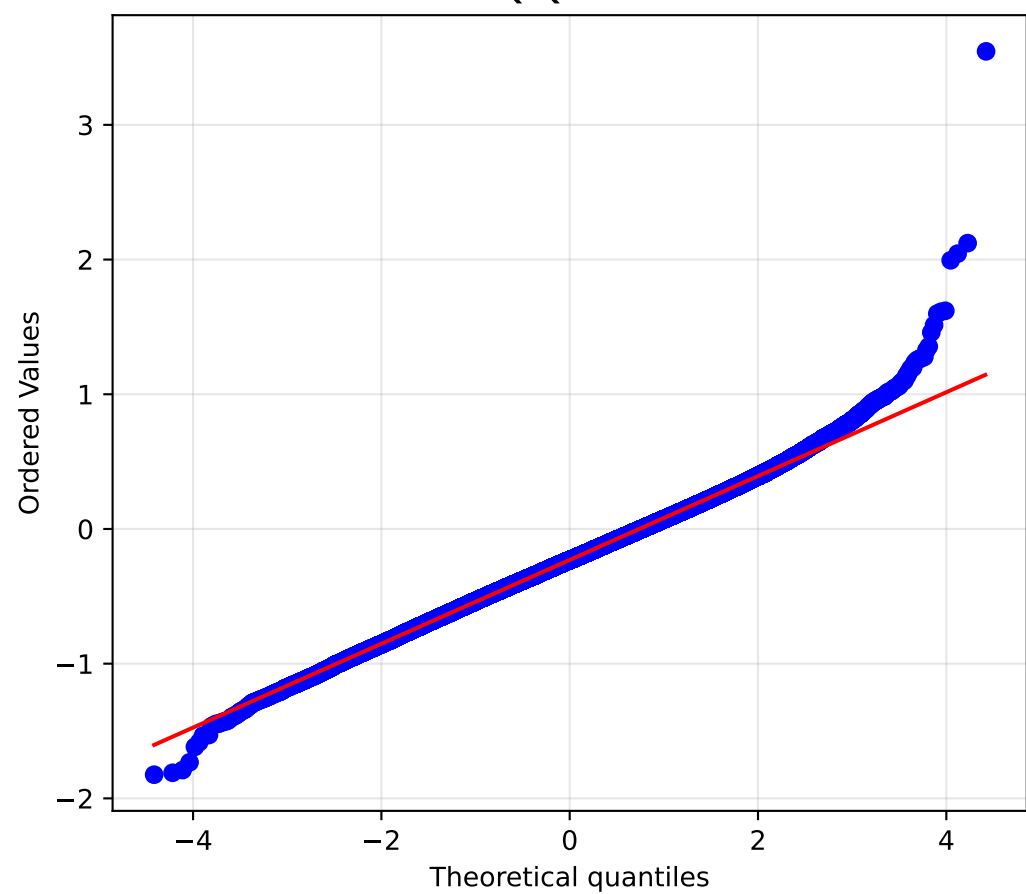


AMPERE Autocorrelation (Lat 75-85°, MLT 15-21)

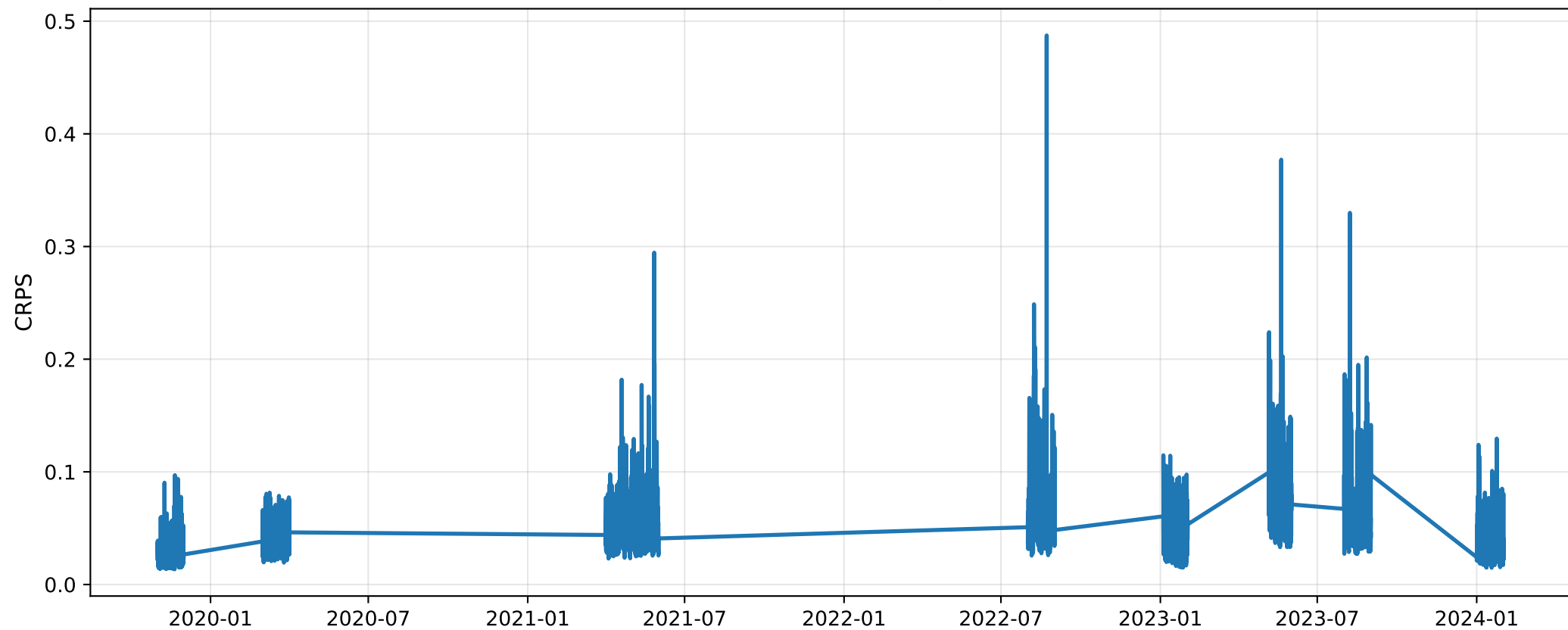


Predicted Autocorrelation (Lat 75-85°, MLT 15-21)

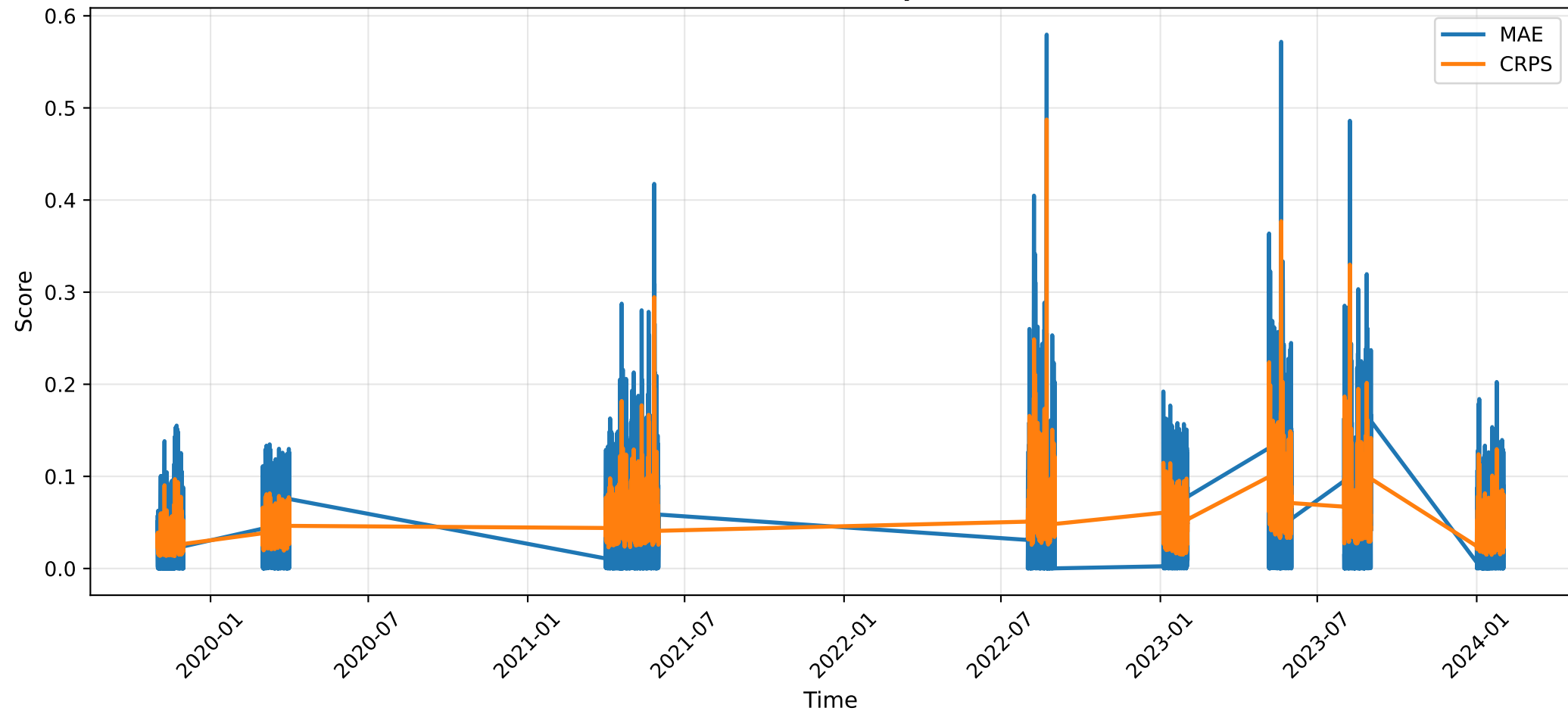


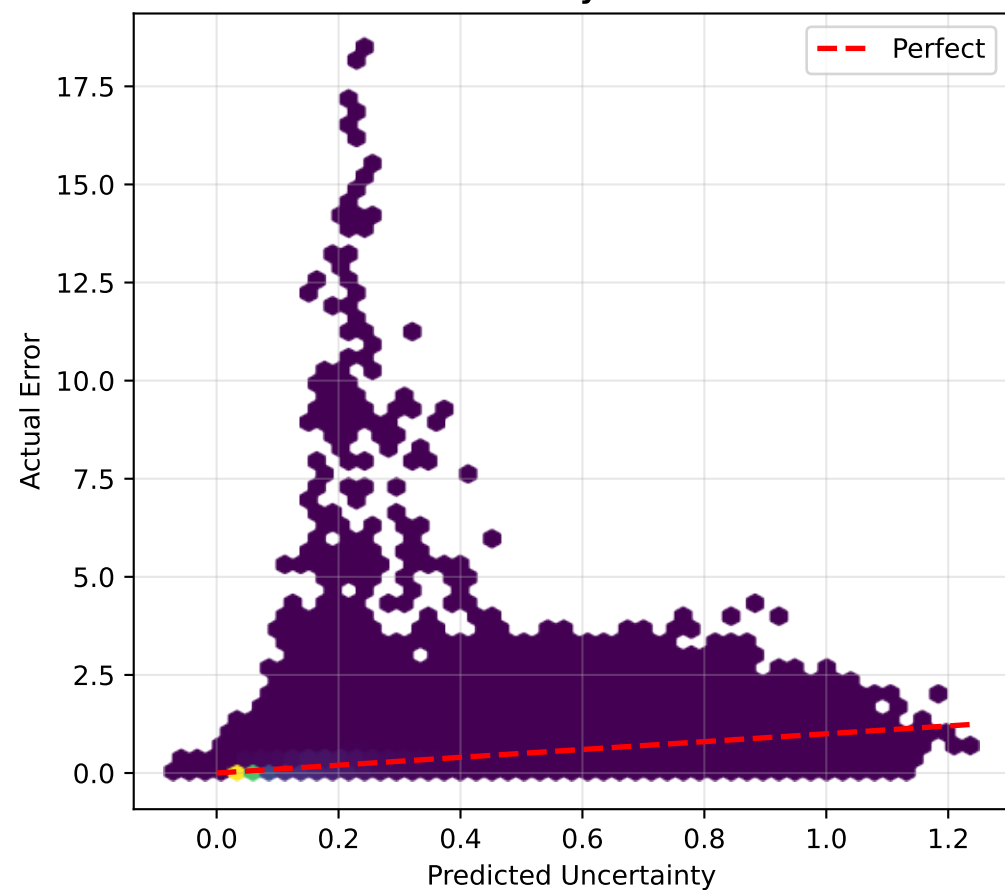
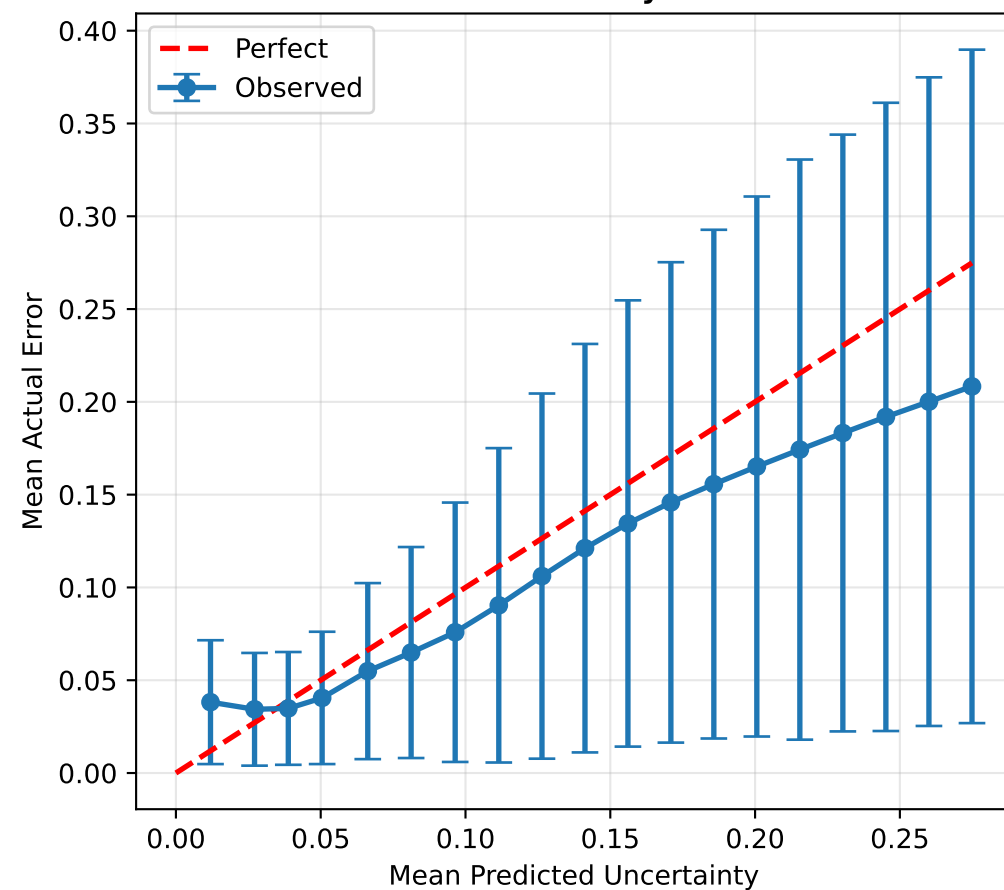
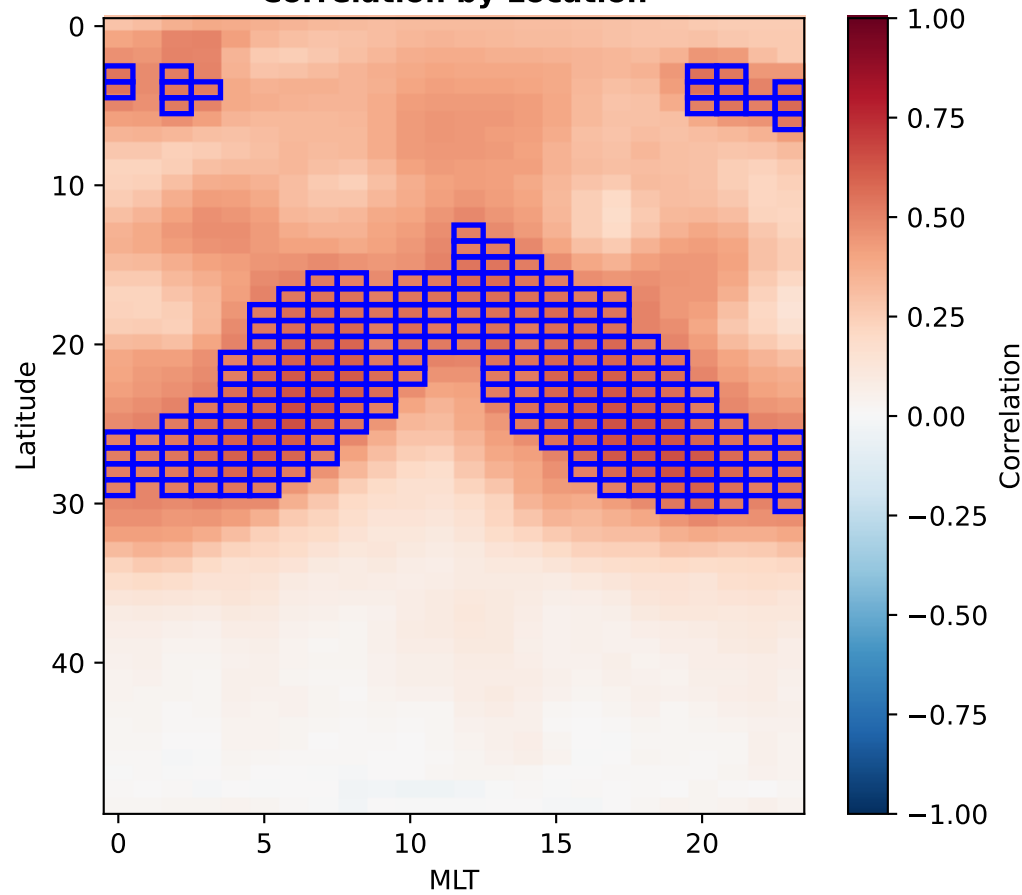
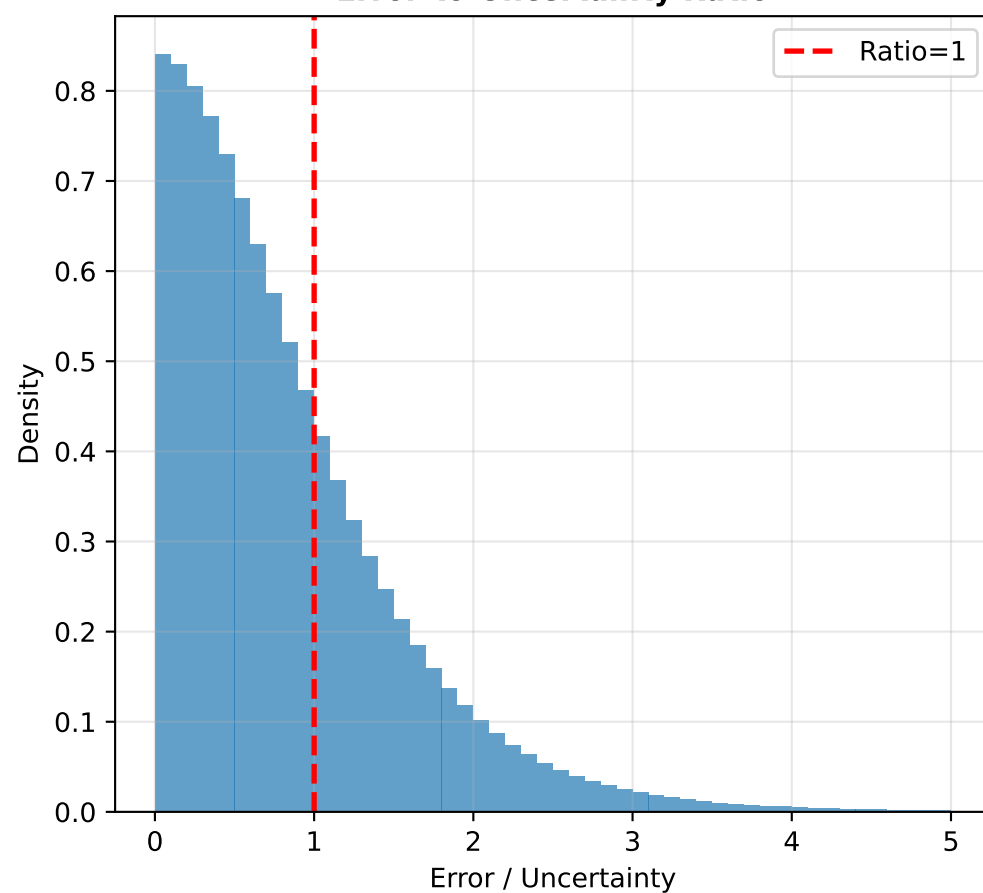
Calibration: Uncertainty vs Error**Z-Score Distribution****Calibration Curve****Q-Q Plot**

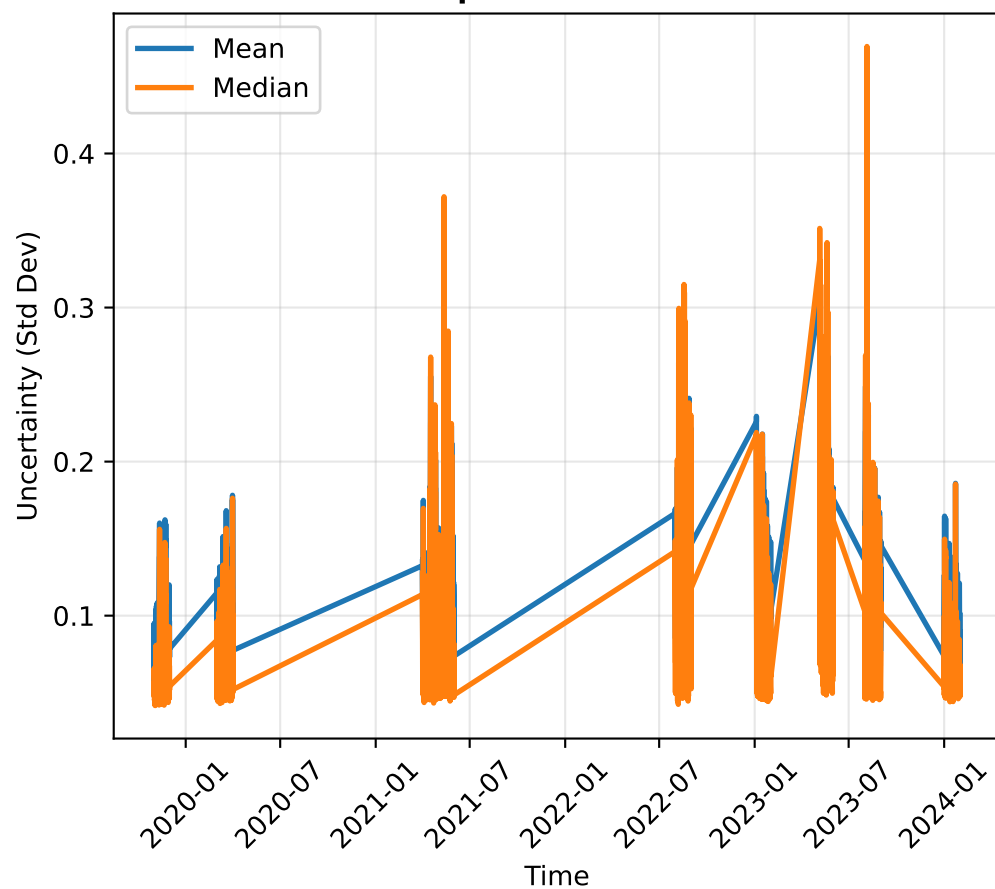
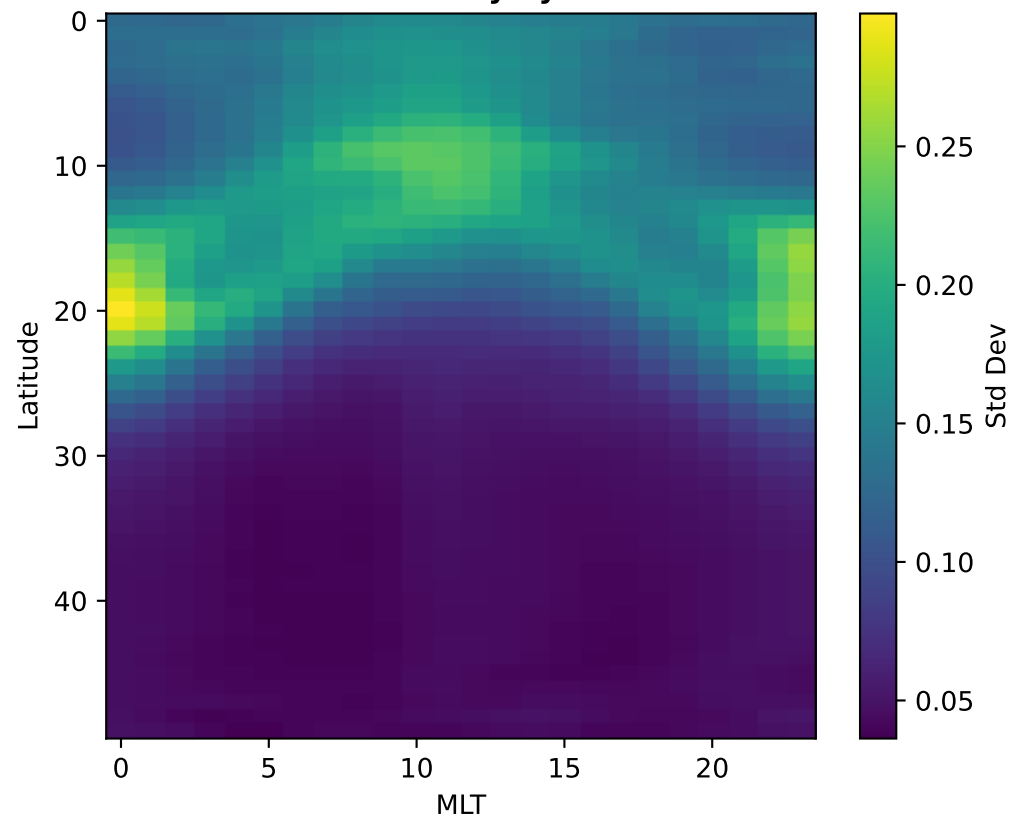
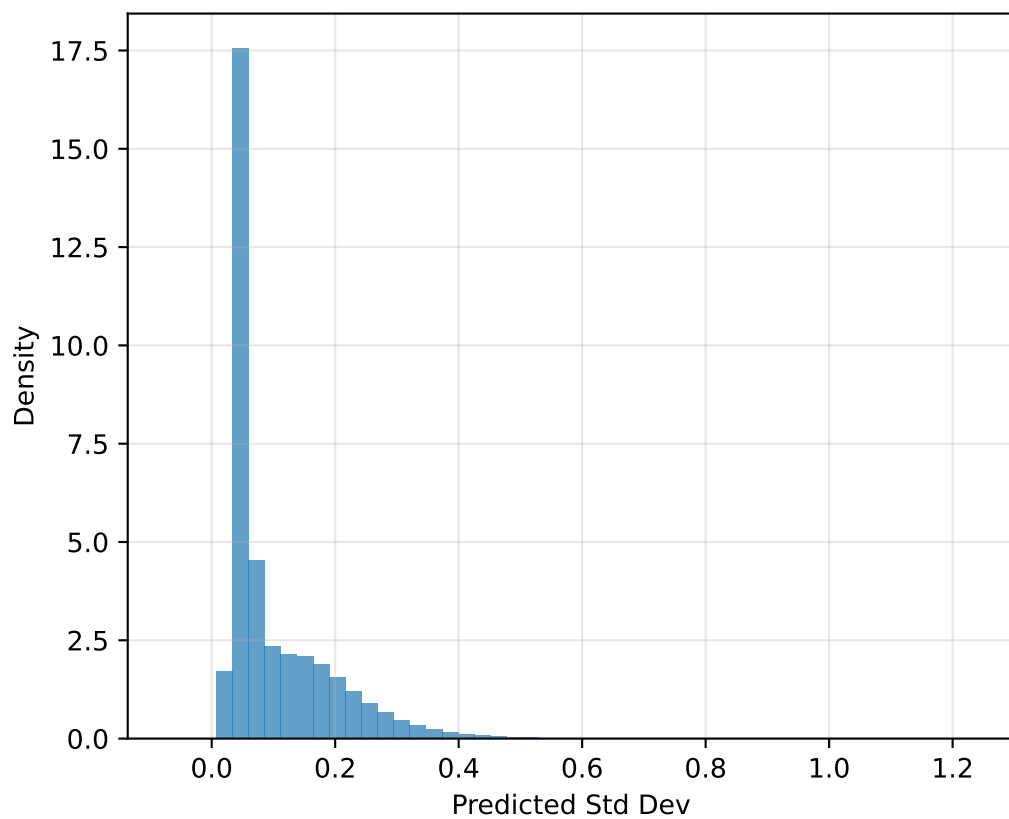
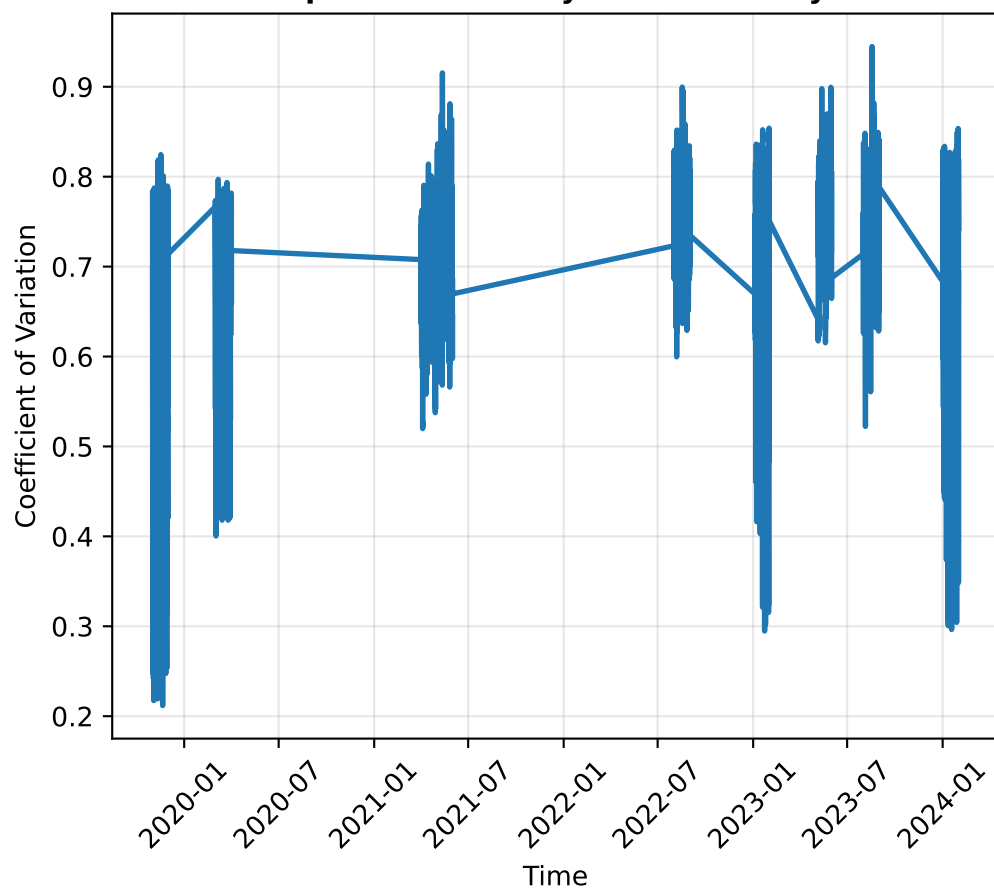
CRPS Over Time (Lat 75-85°, MLT 15-21)



MAE vs CRPS Comparison



Uncertainty vs Error**Binned Analysis****Correlation by Location****Error-to-Uncertainty Ratio**

Sharpness Over Time**Mean Uncertainty by Location****Distribution of Uncertainties****Spatial Variability of Uncertainty**

Numerical Metrics Summary

Deterministic Performance:

Point Correlation: 0.694

Point RMSE: 0.062

Point MAE: 0.047

Point Bias: 0.034

Uncertainty-Error Corr: 0.418

Optimal Lag: -1 timesteps

Max Cross-Correlation: 0.696

Uncertainty Quantification:

Coverage within 1σ : 99.3% (target: 68.3%)

Coverage within 2σ : 100.0% (target: 95.4%)

Z-score mean: -0.230 (target: 0)

Z-score std: 0.311 (target: 1)

Mean CRPS: 0.0437

CRPS/MAE Ratio: 0.932

Uncertainty-Error Correlation: 0.565

Median Error/Uncertainty Ratio: 0.655

Executive Summary

Point Performance:

Point prediction at Lat 75-85°, MLT 15-21 shows moderate correlation ($r=0.694$) with $RMSE=0.062$. Model exhibits positive bias of 0.034. Uncertainty-error correlation is 0.418, indicating moderate awareness of prediction quality.

Spatial Performance:

Spatial analysis at MLT 15-21 shows mean $RMSE$ of 0.076 across all latitudes. Maximum errors (0.174) occur at high latitudes.

Temporal Alignment:

Model leads observations by 1 timesteps (anticipates changes). Maximum achievable correlation is 0.696.

Temporal Dynamics:

Autocorrelation analysis shows decorrelation timescales of 20 timesteps for observations and 20 timesteps for predictions. Model captures temporal persistence well.

Uncertainty Quality:

Uncertainty calibration is poor: 99.3% within 1σ (expected 68.3%), 100.0% within 2σ (expected 95.4%). Model is underconfident (uncertainties too wide). Z-score statistics: mean=-0.230 (ideal: 0), std=0.311 (ideal: 1).

$CRPS/MAE = 0.93$; Uncertainty reduces error by 6.8% compared to just using the mean. Mean $CRPS=0.0437$, Mean $MAE=0.0469$.

Uncertainty-error correlation is 0.565, indicating weak relationship. Median error/uncertainty ratio is 0.655 (underconfident (uncertainties too large)). Model shows moderate awareness of uncertainty patterns.

Uncertainty Characteristics:

Model sharpness: mean uncertainty=0.1025, median=0.0594. Temporal variability ($CV=0.364$) suggests condition-dependent uncertainty. Spatial variability ($CV=0.604$) indicates location-dependent confidence.